

A Fuzzy Time Series-Based Neural Network Approach to Option Price Forecasting

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Abstract. Recently, option price forecasting has become a popular financial issue. Being affected by many factors, option price forecasting remains a challenging problem. This paper proposes a new method to forecast the option price. The proposed method, termed as fuzzy time series-based neural network (FTSNN), is a hybrid method composed of a fuzzy time series model and a neural network model. In FTSNN, the fuzzy time series model is used to select a dataset for training the neural network model for prediction. The experiment results show that FTSNN outperforms several existing methods in terms of MAE and RMSE.

Keywords: Option Price Forecasting, Hybrid method, Fuzzy Time Series, Neural Networks.

1 Introduction

The option is an important tool for risk management in financial markets [1]. For example, a producer can buy a put option to prevent a profit loss due to the decrease of the price of his product in the future. Similarly, a customer can buy a call option to buy his desired product at an expected price in the future. However, an option is like an insurance policy in that one has to pay premium for an option. The premium, also called the price, of an option is determined by many factors such as the current stock price, the option strike price, the time to expiration, the volatility of the stock price and the risk-free interest rates [2]. Being affected by many factors, option price forecasting remains a challenging problem. Due to the popularity of using options as a risk management tool, the option price forecasting becomes an important task. The well-known Black-Scholes model (B-S model) was introduced in 1973 [2] for option pricing. However, the B-S model is limited due to the fact that many of its assumptions are not true in a real life financial market. For this reason, many researches proposed new methods to predict the option price.

Artificial neural networks (ANNs) model is popular for financial forecasting, such as the stock price indices forecasting [3], [4] and the exchange rate forecasting [5], [6]. The neural network model is also widely used in option price forecasting. For example,

many researchers proposed hybrid neural network models to predict the option price [1], [7], [8], [9]. Recently, the fuzzy time series model is gaining its popularity in financial forecasting, such as the stock price indices forecasting [10], [11], the exchange rate forecasting [12], and the futures exchange index forecasting [13]. In this paper, we proposed a new method to predict the option price of “Taiwan Stock Exchange Stock Price Index Options (TXO)”. Our proposed method, termed as fuzzy time series-based neural network (FTSNN), is a hybrid method composed of a fuzzy time series model and a neural network model. In FTSNN, the fuzzy time series model is used to select a dataset for training the neural network model for prediction.

The remainder of this paper is organized as follows. Section 2 reviews the definitions of fuzzy time series. Section 3 introduces the procedure of FTSNN. Section 4 describes an application of the FTSNN to option price forecasting. Section 5 compares the performance of the FTSNN with other proposed methods. Section 6 gives the conclusions of this paper.

2 Fuzzy Time Series

This paper proposes to combine a two-factor fuzzy time series and a neural network for option price forecasting. In the following, we will briefly review the definition of fuzzy time series.

According to [13], [14], [15], the following definitions are given to fuzzy time series model.

Definition 1: Let $Y(t)$ ($t = \dots, 0, 1, 2, \dots$), a subset of R^1 , be the universe of discourse on which fuzzy sets $f_i(t)$ ($i = 1, 2, \dots$) are defined. If $F(t)$ is a collection of $f_i(t)$. Then $F(t)$ is called a fuzzy time series defined on $Y(t)$.

Definition 2: If for any $f_j(t) \in F(t)$, there exists an $f_i(t-1) \in F(t-1)$, such that there exists a fuzzy relation $R_{ij}(t, t-1)$ and $f_j(t) = f_i(t-1) \circ R_{ij}(t, t-1)$ where ‘ $\circ$ ’ is the max-min composition, then $F(t)$ is said to be caused by $F(t-1)$ only, and we denote this as $F(t-1) \rightarrow F(t)$.

Definition 3: If $F(t)$ is caused by $F(t-1)$, $F(t-2)$, ..., and $F(t-n)$, $F(t)$ is called a one-factor n -order fuzzy time series, and is denoted by

$$F(t-n), \dots, F(t-2), F(t-1) \rightarrow F(t).$$

Definition 4: If $F(t)$ is caused by $(F_1(t-1), F_2(t-1))$, $(F_1(t-2), F_2(t-2))$, ..., $(F_1(t-n), F_2(t-n))$, $F(t)$ is called a two-factor n -order fuzzy time series, which is denoted by

$$(F_1(t-n), F_2(t-n)), \dots, (F_1(t-2), F_2(t-2)), (F_1(t-1), F_2(t-1)) \rightarrow F(t).$$

Let $F_1(t) = X_t$ and $F_2(t) = Y_t$, where X_t and Y_t are fuzzy variables whose values are possible fuzzy sets of the first factor and the second factor, respectively, on day t . Then, a two-factor n -order fuzzy logic relationship (FLR) [14] can be expressed as:

$$(X_{t-n}, Y_{t-n}), \dots, (X_{t-2}, Y_{t-2}), (X_{t-1}, Y_{t-1}) \rightarrow X_t,$$

where $(X_{t-n}, Y_{t-n}), \dots, (X_{t-2}, Y_{t-2})$ and (X_{t-1}, Y_{t-1}) , are referred to as the left-hand side (LHS) of the relationship, and X_t is referred to as the right-hand side (RHS) of the relationship.

3 The FTSNN Method

When there are many different fuzzy sets at the LHS of a FLR, it is difficult to find a matched FLR for prediction. Some measures have to be taken to counter this problem [12]. In this paper, we use the fuzzy time series to select several similar FLRs for prediction, and use the neural network to construct a prediction model. The proposed method is a hybrid method in the sense that the fuzzy time series model is used to select training data set (i.e., FLRs) and the neural network is used to build a prediction model using the selected training dataset. The detail procedure of FTSNN method is described in the following.

Step 1: Divide the universe of discourse

The universe of discourse of the first factor is defined as $U = [D_{\min} - D_1, D_{\max} + D_2]$, where $D_{\min}$ and $D_{\max}$ are the minimum and maximum of the first factor, respectively; D_1 and D_2 are two positive real numbers to divide the universe of discourse into n equal length intervals. The universe of discourse of the second factor is defined as $V = [V_{\min} - V_1, V_{\max} + V_2]$, where $V_{\min}$ and $V_{\max}$ are the minimum and maximum of the second factor, respectively; similarly, V_1 and V_2 are two positive real numbers used to divide the universe of discourse of the second factor into m equal length intervals. Note that the length of the interval of each factor is determined by its largest value of the factor in historical data.

Step 2: Define fuzzy sets

Linguistic terms A_i , $1 \leq i \leq n$, are defined as fuzzy sets on the intervals of the first factor. They are defined as follows:

$$\begin{aligned} A_1 &= 1/u_1 + 0.5/u_2 + 0/u_3 + \dots + 0/u_{n-2} + 0/u_{n-1} + 0/u_n, \\ A_2 &= 0.5/u_1 + 1/u_2 + 0.5/u_3 + \dots + 0/u_{n-2} + 0/u_{n-1} + 0/u_n, \\ &\vdots \\ A_{n-1} &= 0/u_1 + 0/u_2 + 0/u_3 + \dots + 0.5/u_{n-2} + 1/u_{n-1} + 0.5/u_n, \\ A_n &= 0/u_1 + 0/u_2 + 0/u_3 + \dots + 0/u_{n-2} + 0.5/u_{n-1} + 1/u_n, \end{aligned}$$

where u_i denotes the i^{th} interval of the first factor.

Similarly, linguistic term B_j , $1 \leq j \leq m$, is defined as a fuzzy set on the intervals of the second factor. They are defined as follows:

$$\begin{aligned} B_1 &= 1/v_1 + 0.5/v_2 + 0/v_3 + \dots + 0/v_{m-2} + 0/v_{m-1} + 0/v_m, \\ B_2 &= 0.5/v_1 + 1/v_2 + 0.5/v_3 + \dots + 0/v_{m-2} + 0/v_{m-1} + 0/v_m, \\ &\vdots \\ B_{m-1} &= 0/v_1 + 0/v_2 + 0/v_3 + \dots + 0.5/v_{m-2} + 1/v_{m-1} + 0.5/v_m, \\ B_m &= 0/v_1 + 0/v_2 + 0/v_3 + \dots + 0/v_{m-2} + 0.5/v_{m-1} + 1/v_m, \end{aligned}$$

where v_i is the i^{th} intervals of the second factor.

Step 3: Determine the order of FTSNN

FTSNN uses a two-factor n -order fuzzy time series model to search for similar FLRs for building a neural network prediction model. Similar FLRs imply similar trends in the historical data. The order of FTSNN can be regarded as the length of the trends. How to determine a suitable length of the trends is a problem to the FTSNN method. In this paper, we choose to set the order from 1 to 5 and build five different prediction models. Then, we choose the one with the best prediction accuracy as the final prediction model. Note that since n can be equal to any of 1, 2, 3, 4, or 5, step a to step d in the following will be performed five times to build five different models.

(a) Construct FLRs database

For the historical data on day i , let X_{i-n} , Y_{i-n} denote the fuzzy set of $F_1(i-n)$ and $F_2(i-n)$ of the fuzzy time series. Let X_i denotes the fuzzy set of $F_1(i)$. The FLRs database on day i can be represented as follows:

$$(X_{i-n}, Y_{i-n}), \dots, (X_{i-2}, Y_{i-2}), (X_{i-1}, Y_{i-1}) \rightarrow X_i.$$

(b) Construct the LHS of FLR of the predicting day (assume that day t is the predicting day)

The LHS of the FLR on day t can be represented as follows:

$$(X_{t-n}, Y_{t-n}), \dots, (X_{t-2}, Y_{t-2}), (X_{t-1}, Y_{t-1}).$$

(c) Search for similar FLRs

Calculate the Euclidean distance (ED) of the LHS of the FLR on day t against the LHS of each candidate FLR in the FLRs database. Then, we select the top five FLRs with the smallest Euclidean distance from the FLR database as the training example for building a neural network. The Euclidean distance between the FLR on day t and the FLR on day i can be calculated according to Formula 1 2, and 3.

$$ED_{Ai} = \sqrt{(IX_{t-n} - RX_{i-n})^2 + \dots + (IX_{t-2} - RX_{i-2})^2 + (IX_{t-1} - RX_{i-1})^2}, \quad (1)$$

$$ED_{Bi} = \sqrt{(IY_{t-n} - RY_{i-n})^2 + \dots + (IY_{t-2} - RY_{i-2})^2 + (IY_{t-1} - RY_{i-1})^2}, \quad (2)$$

$$ED_i = (ED_{Ai} + ED_{Bi}) / 2. \quad (3)$$

In the above formulae, IX_{t-n} and IY_{t-n} are the subscripts of the fuzzy sets of the first factor and the second factor, respectively, of the LHS of day t 's FLR. Similarly, RX_{i-n} and RY_{i-n} are subscripts of the first factor and the second factor, respectively, of the LHS of day i 's FLR.

(d) Build neural network models

With the top five similar FLRs as the training examples, we can train a radial basis function neural network (RBFNN) model. The training framework of the neural network model is shown in Figure 1.

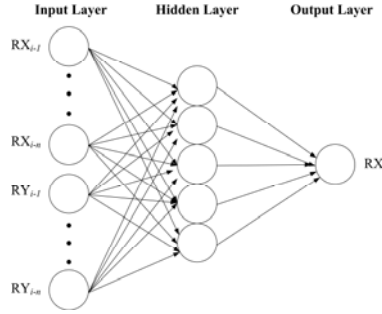


Fig. 1. The training framework of the neural network model

(e) Model selection

Once the five models are built, we can choose the best model for prediction. In this paper, we use the prediction accuracy on day $t-1$ as the criterion of model selection. In other words, we feed the LHS of the FLR on day $t-1$ as the input of a trained model and calculate the error according to Formula 4. We then choose the model with the smallest error. Note that in Formula 4, the *forecasted RHS* denotes the subscript of the forecasted fuzzy set on day $t-1$, and the *testing RHS* denotes the actual subscript of the fuzzy set of the RHS on day $t-1$.

$$\text{Error function} = | \text{Forecasted RHS} - \text{Testing RHS} | \quad (4)$$

Step 4: Forecasting

When the best n -order FTSNN model is constructed in step 3, we can perform forecasting by feeding the LHS of the FLR on the predicting day into the neural network to get the forecasted subscript of the RHS on the predicting day.

Because the forecasted value is a subscript of a fuzzy set, we have to defuzzify it into the option price. We use the weighted average method as the defuzzification method. Formula 5 shows the weighted average method.

$$\text{forecast_value} = \begin{cases} \frac{M[1] + 0.5 \cdot M[2]}{1 + 0.5} & k = 1, \\ \frac{0.5 \cdot M[k-1] + M[k] + 0.5 \cdot M[k+1]}{0.5 + 1 + 0.5} & 1 < k \leq n-1, \\ \frac{0.5 \cdot M[n-1] + M[n]}{0.5 + 1} & k = n, \end{cases} \quad (5)$$

where $M[k]$ denotes the midpoint value of the fuzzy set k . Note that an iteration of the above procedure (step 1 through step 4) predicts only one forecasting value.

4 Option Price Forecasting Using FTSNN

To forecast the price of “Taiwan Stock Exchange Stock Price Index Option (TXO)”, we choose the closing price of TXO as the first factor in FTSNN and the “Taiwan Stock

Exchange Capitalization Weighted Stock Index (TAIEX)” as the second factor. A part of the historical data is shown in Table 1. With this historical data, U is set to [0, 1000] and is divided into 100 intervals. That is, $u_1=[0, 10]$, $u_2=[10, 20]$, ..., and $u_{100}=[990, 1000]$. For the second factor, V is set to [3000, 13000] and is divided into 240 intervals. That is, $v_1=[3000, 3050]$, $v_2=[3050, 3100]$, ..., and $v_{240}=[12950, 13000]$. Having defined the intervals, we fuzzify the historical data into fuzzy sets. Table 2 shows parts of the fuzzified historical data. Then, we construct the FLRs database from the fuzzified historical data. A 2-order FLR database for this historical data is showed in Table 3.

Having constructed the FLR database, we can forecast the option price. For example, if we want to forecast the option price on day 11, we first construct the LHS of FLR on day 11 as follows,

$$(A_{25}B_{117}), \quad (A_{30}B_{117}).$$

Then, we calculate the Euclidean distance between LHS of the FLR on day 11 and the LHS of each candidate FLRs in the FLRs database. Table 4 shows the calculated Euclidean distances. Then, we select top five similar FLRs from the FLR database. In this example, FLR₈, FLR₅, FLR₇, FLR₆, FLR₄ are selected. Finally, we use these FLRs as training examples to build a neural network model.

Table 1. Parts of historical data

Date	Day 1	Day 2	Day 3	Day 4	Day 5
Option Prices	585	481	418	420	368
Stock indices	6143.12	6060.46	5988.37	5982.12	5935.99
Date	Day 6	Day 7	Day 8	Day 9	Day 10
Option Price	376	398	286	255	304
Stock indices	5942.85	5975.66	5879.08	5853.94	5889.52

Table 2. Parts of fuzzified dataset

Day 1	Day 2	Day 3	Day 4	Day 5
(A ₅₈ , B ₁₂₂)	(A ₄₈ , B ₁₂₁)	(A ₄₁ , B ₁₁₉)	(A ₄₂ , B ₁₁₉)	(A ₃₆ , B ₁₁₈)
Day 6	Day 7	Day 8	Day 9	Day 10
(A ₃₇ , B ₁₁₈)	(A ₃₉ , B ₁₁₉)	(A ₂₈ , B ₁₁₇)	(A ₂₅ , B ₁₁₇)	(A ₃₀ , B ₁₁₇)

To select a model for forecasting, we use FLR₈, which is the FLR on day 10, as the testing data. To perform the testing, we feed the LHS of the FLR on day 10 into the neural network to get the predicted subscript of the fuzzy set of the RHS on day 10. Then, we calculate the error between the predicted subscript and the actual subscript. For this example, the forecasted subscript is 27 and the actual subscript is 30. Hence, the error is equal to 3.Note that the procedure above will be performed on 1-order, 2-order, 3-order, 4-order and 5-order fuzzy time series models. The one with the smallest error is the model for forecasting.

Table 3. The FLRs database

FLR ₁	(A ₅₈ , B ₁₂₂)	(A ₄₈ , B ₁₂₁)	→	A ₄₁
FLR ₂	(A ₄₈ , B ₁₂₁)	(A ₄₁ , B ₁₁₉)	→	A ₄₂
...	...	...	...	...
FLR ₈	(A ₂₈ , B ₁₁₇)	(A ₂₅ , B ₁₁₇)	→	A ₃₀

Table 4. Euclidean distances

	ED_A	ED_B	ED
FLR ₁	$\sqrt{(25-58)^2 + (30-48)^2}$	$\sqrt{(117-122)^2 + (117-121)^2}$	750.9726
FLR ₂	$\sqrt{(25-48)^2 + (30-41)^2}$	$\sqrt{(117-121)^2 + (117-119)^2}$	349.1129
...	...	...	...
FLR ₈	$\sqrt{(25-28)^2 + (30-25)^2}$	$\sqrt{(117-117)^2 + (117-117)^2}$	16.7772

For the above example, assume that a 2-order neural network model is selected, and the forecasted subscript is 35 on day 11. Substituting 345, 355 and 365 for $M[34]$, $M[35]$, and $M[36]$, respectively in Formula 5, we get 355 as the forecasted option price on day 11. Note that the actual option price on day 11 is 360 in this example.

5 Results and Performance

5.1 Dataset

The dataset of this paper are the daily transaction data of TXO and TAIEX from January 3, 2005 to December 29, 2006. This paper investigates a sample of 23,819 call option price data. Call options can be divided into three categories according to their S/K ratio. The distribution of the dataset is shown in Table 5. We refer to [8] and [9] for the definition of the categories. Our dataset comprises 30 different strike price from 5,200 to 8,200 and 12 different expiration dates from January 2005 to December 2006.

Table 5. Data distribution according to moneyness

Catogeries	Moneyness	Number
In-the-money	$S/K > 1.02$	8938
At-the-money	$0.95 < S/K \leq 1.02$	7508
Out-of-the-money	$S/K \leq 0.95$	7373

Notes: S is the spot price; K is the strike price.

Note that the option prices of the beginning 10 transaction dates of each option are not predicted due to insufficient historical data. In predicting the option price of a specific date, the option prices of the previous transaction dates become the historical data.

5.2 Performance Measures

Two different performance measures, mean absolute error (MAE) and root mean square error (RMSE), are used to measure the forecasting accuracy of FTSNN. The formulae are shown as follows:

$$MAE = \frac{\sum_{t=1}^n |A_t - P_t|}{n}, \tag{6}$$

$$RMSE = \sqrt{\frac{\sum_{t=1}^n (A_t - P_t)^2}{n}}, \tag{7}$$

where A_t and P_t denote the real option price and the forecasting option price on day t , respectively.

5.3 Performance

The performance of FTSNN is compared with the other proposed methods which were published in the previous literatures [8] and [9]. Table 6 shows parts of the results of the option price forecasting by FTSNN. The forecasting prices are close to the real option

Table 6. Parts of the results of FTSNN

Strike price	Date	Real option price	FTSNN forecasting
7100	2006/9/25	127	106
7100	2006/9/26	118	118
7100	2006/9/27	132	117
7100	2006/9/28	95	125
7100	2006/9/29	91	94
7100	2006/10/2	119	115
7100	2006/10/3	109	106

Table 7. The performance in RMSE and MAE

RMSE						
Moneyness	FTSNN	GARCH	GJR	Grey-GJR	EGARCH	Grey-EGARCH
In	72.79	85.49	76.19	73.76	73.90	72.11*
At	36.99*	44.02	41.06	40.11	41.35	40.26
Out	16.19*	25.73	25.53	25.89	26.34	26.13
MAE						
Moneyness	FTSNN	GARCH	GJR	Grey-GJR	EGARCH	Grey-EGARCH
In	52.11*	69.54	59.28	56.13	57.02	57.26
At	23.98*	34.73	31.67	30.21	32.17	32.51
Out	9.78*	18.78	17.41	17.26	18.30	18.26

Notes: 1. * denotes the smallest value. 2. GJR denotes GJR–GARCH model. 3. Grey-GJR denotes Grey-GJR–GARCH model.

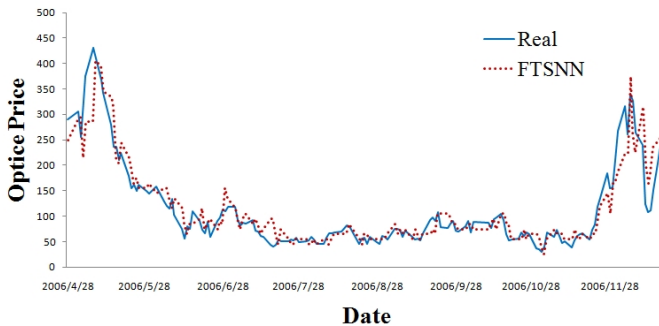


Fig. 2. Time series of the real option price and the forecasting price

prices, except on the dates when the option prices change abruptly. The performance of FTSNN and the other proposed methods are shown in Table 7. Table 7 shows that the performance of a traditional method, GARCH, is always the worst. The other hybrid methods are better than GARCH, especially FTSNN and Grey-EGARCH. In RMSE, the forecasting accuracy of the FTSNN is better than the other methods except the options belonging to in-the-money category. The RMSE of the Grey-EGARCH is better than that of FTSNN, but it is not significant. Table 7 shows that the accuracy of FTSNN is the best in all three categories in MAE. Figure 2 shows the forecasting results of an option with strike price equal to 7,400 and expiration date December 20, 2006.

6 Conclusion

In this paper, we propose the FTSNN method to predict the option price. In FTSNN, we use the fuzzy time series model to select best training data from the historical dataset to train a neural network forecasting model. The experiment results show that FTSNN is more accurate than the other proposed methods in terms of RMSE for options belonging to at-the-money and out-the-money. For in-the-money options, the performance of FTSNN is similar to those of the other methods. Furthermore, in terms of MAE, FTSNN is better than the other methods for all categories of options. Hence, FTSNN offers a useful alternative for option price forecasting.

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